

MobilityStudio: Making MATSim Accessible

A Visual Workbench and a First Step Towards Conversational, AI-Assisted Transport Planning

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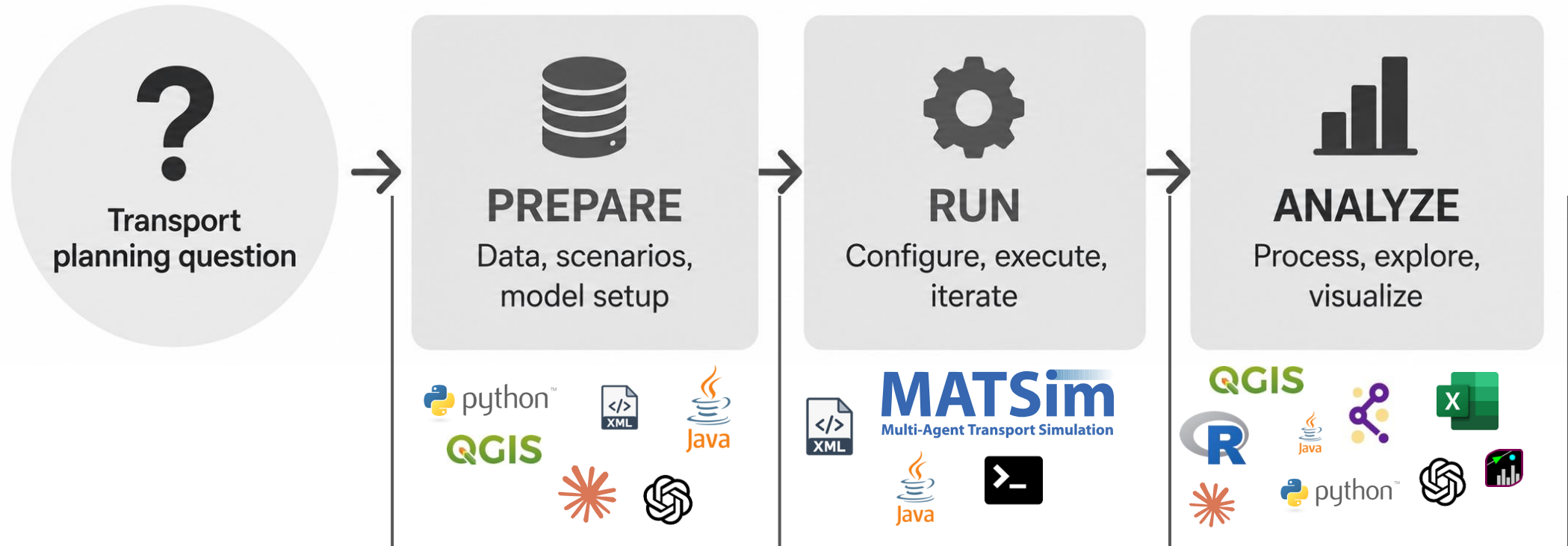


MobilityStudio

senozon

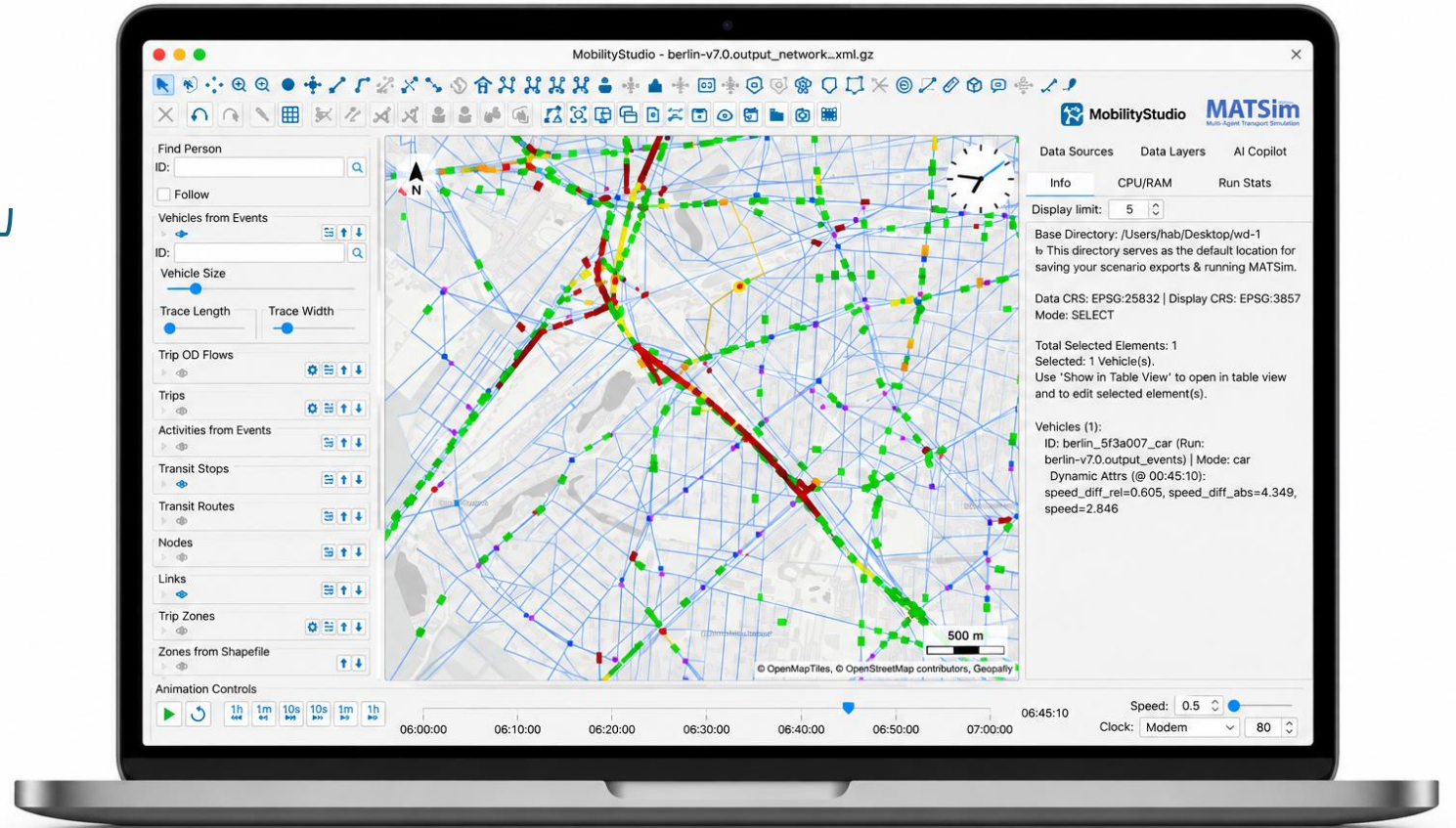
MATSim is powerful, but working with it can still be technical and complicated.

From a transport planning question to a model-based answer typically involves several steps and multiple tools.



? **Can we make this workflow easier?**

We built a visual workbench for MATSim.



Making MATSim accessible: Turning an expert simulation framework into intuitive transport planning software.

Core capabilities.

- A broad set of features, analysis and editing tools.
- Full support for the typical MATSim ecosystem: network, transit schedule, vehicles, facilities, population, counts, DRT, trips, legs, persons from CSV, events, ...



Entire workflow.

Create, run and analyze without (or with) switching between tools and files.



Map-first.

An intuitive GIS-like interface: Layers, selection tools, editing, map interaction.



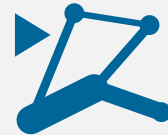
Map + Table.

Two synchronized ways to work with data.



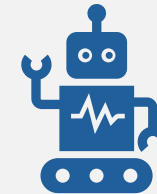
Static Assignment.

A fast path for static network assignment and prototyping.



AI Assistance.

An integrated AI-Copilot in Chat mode and as Agent.

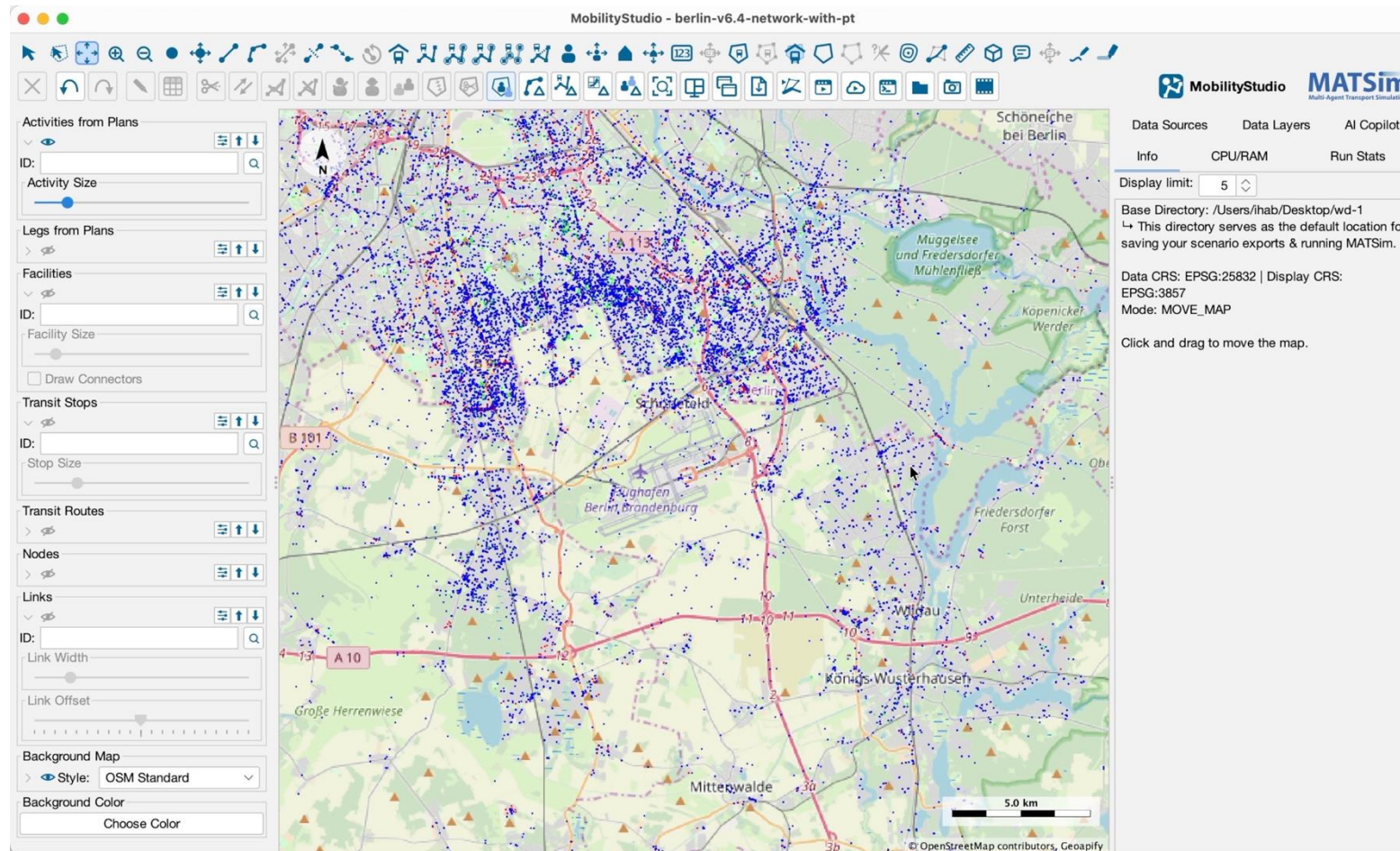


Demo tour.

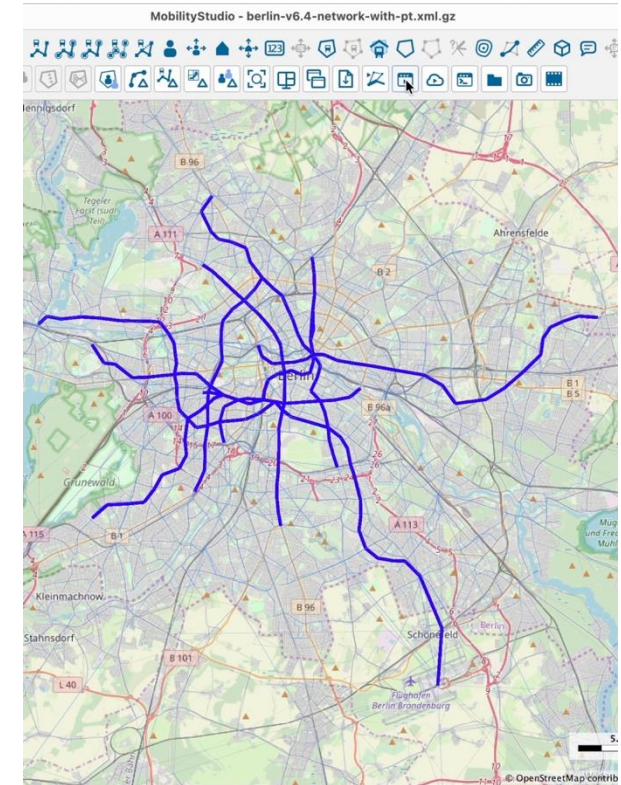
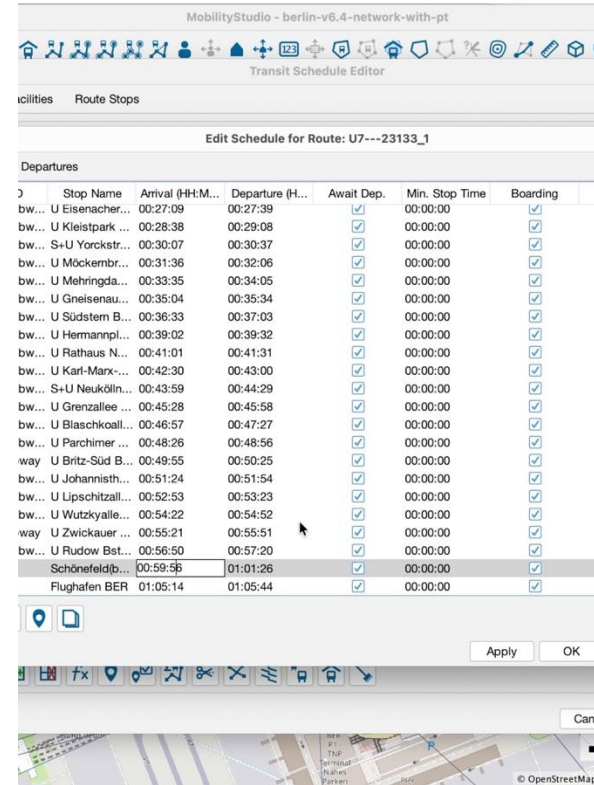
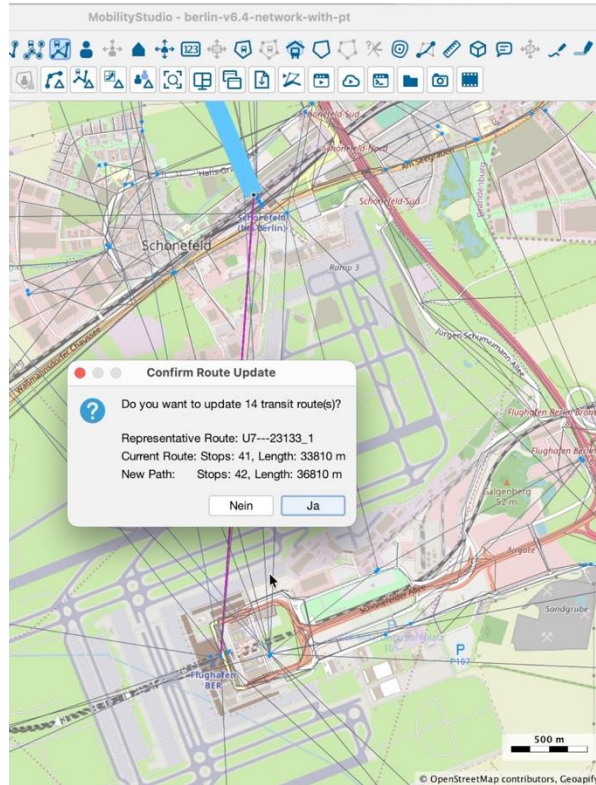
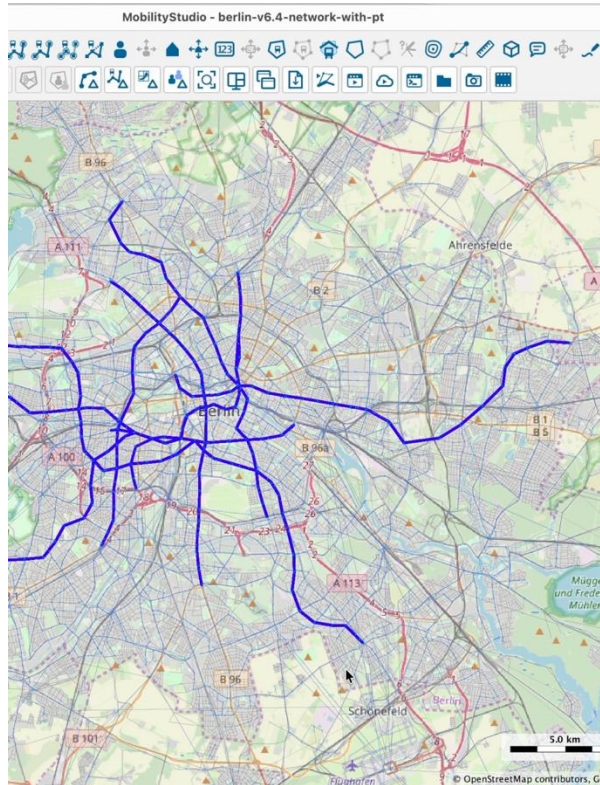
The app starts with a guided demo tour and example data.



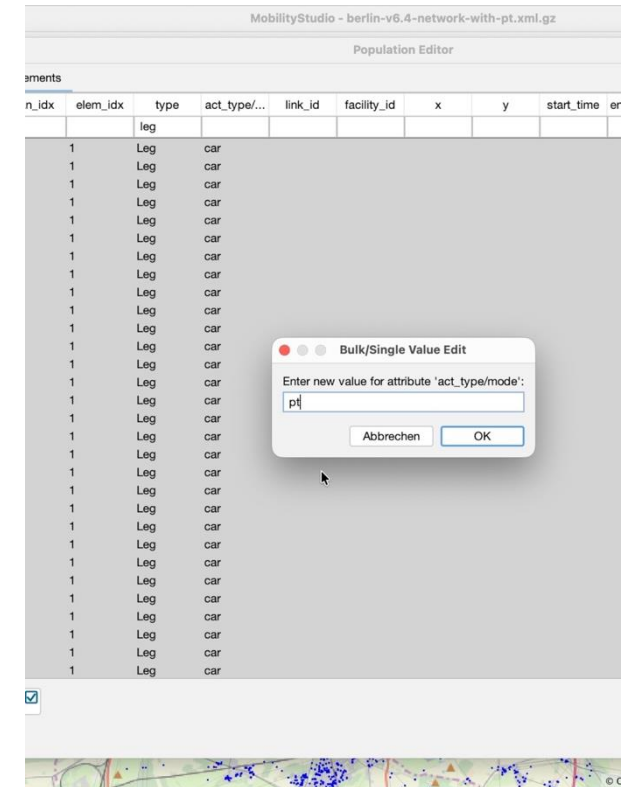
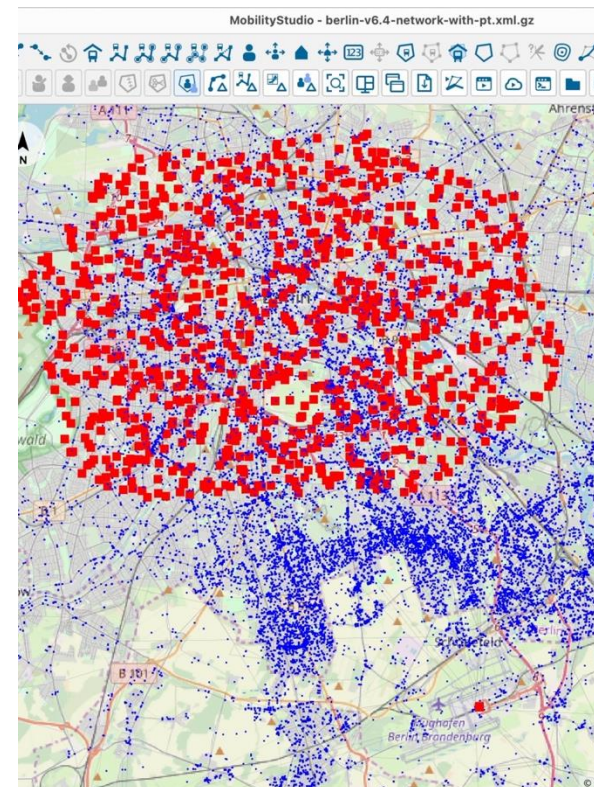
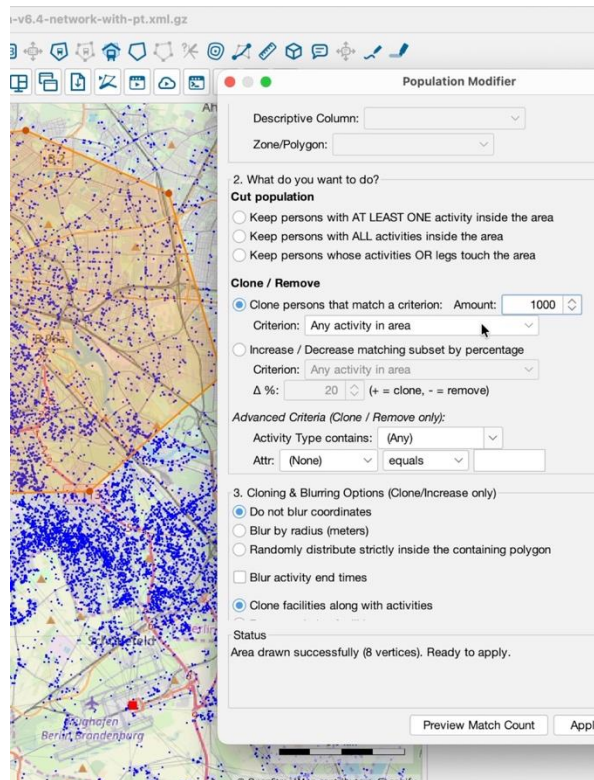
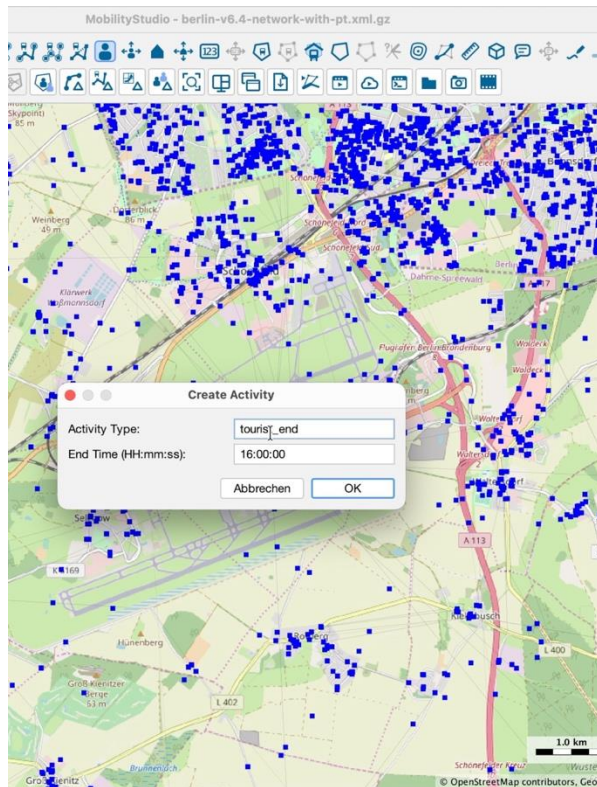
A fictive case study: Extending the subway line U7 in Berlin. Using the Open Berlin Scenario by TU Berlin...



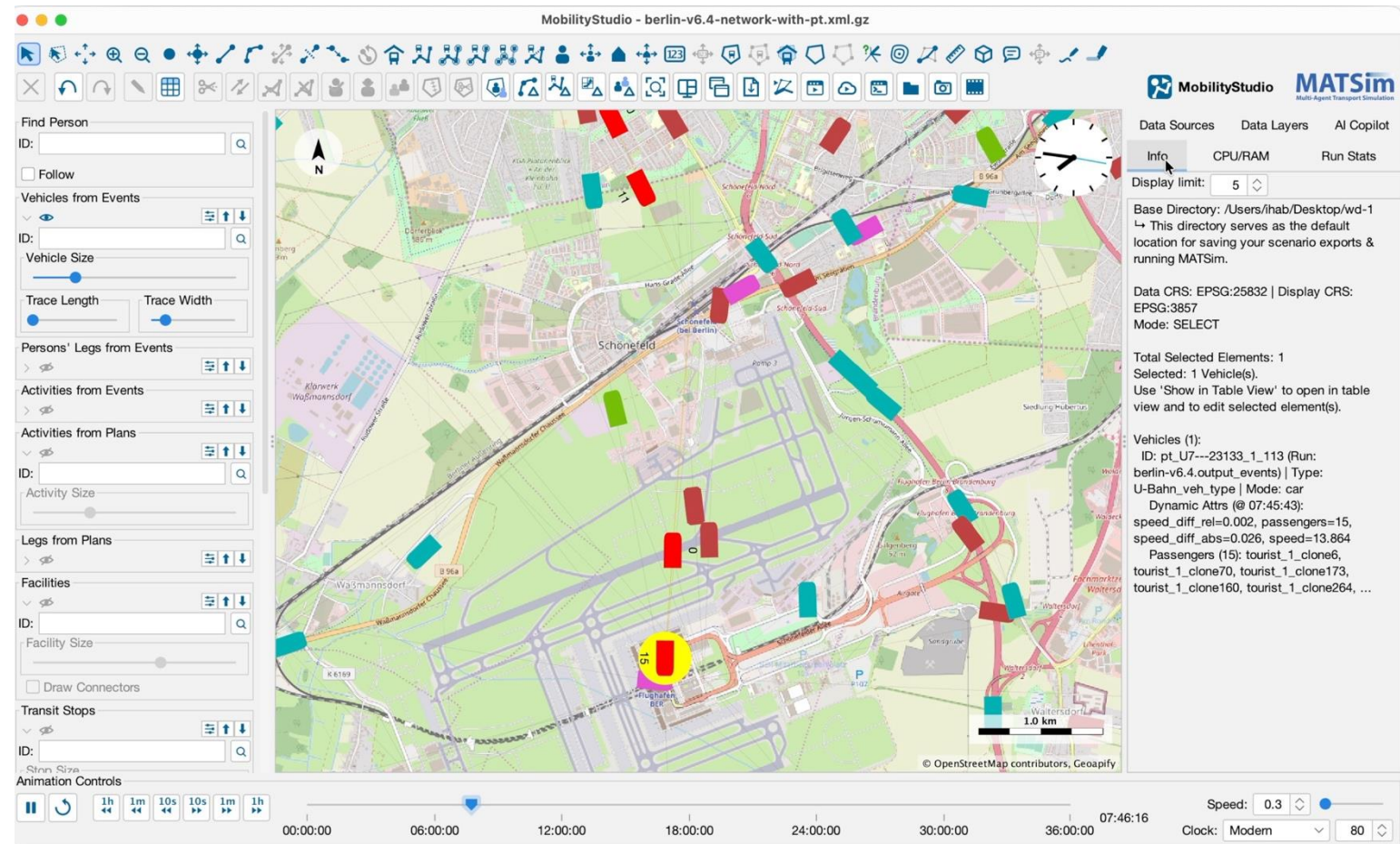
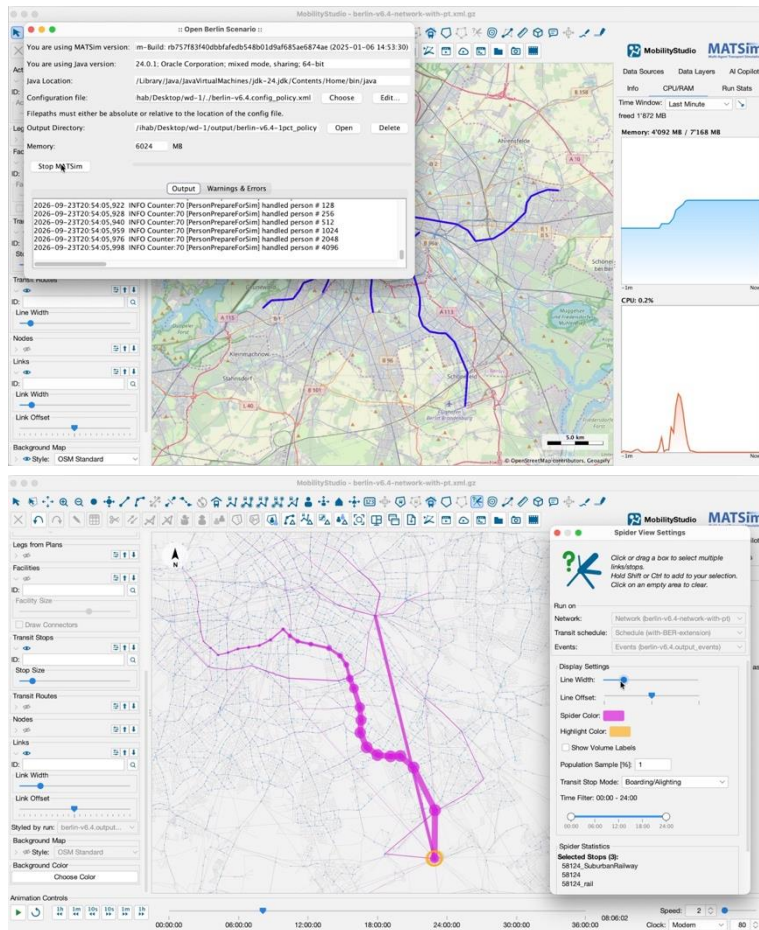
Extend the subway line U7 from Rudow to BER airport. Using interactive map editing features and the table view.



Adding an additional demand segment to the existing population: tourists traveling by public transit to BER.



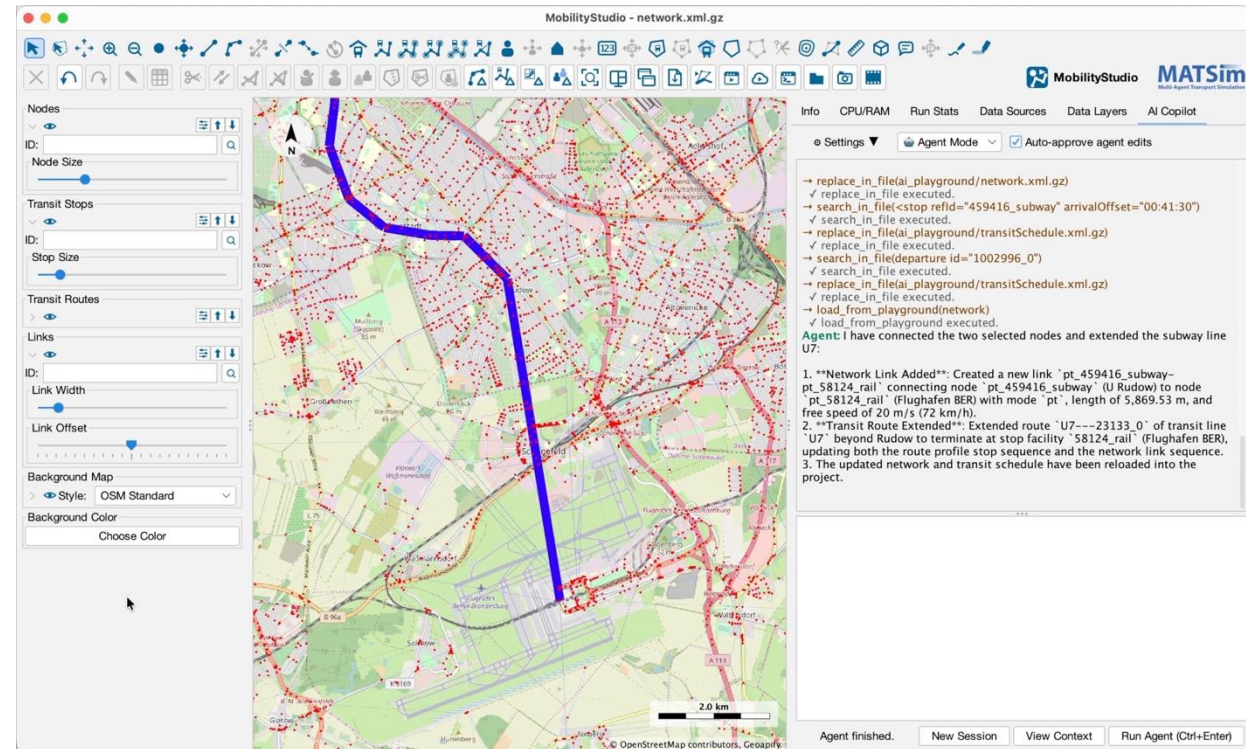
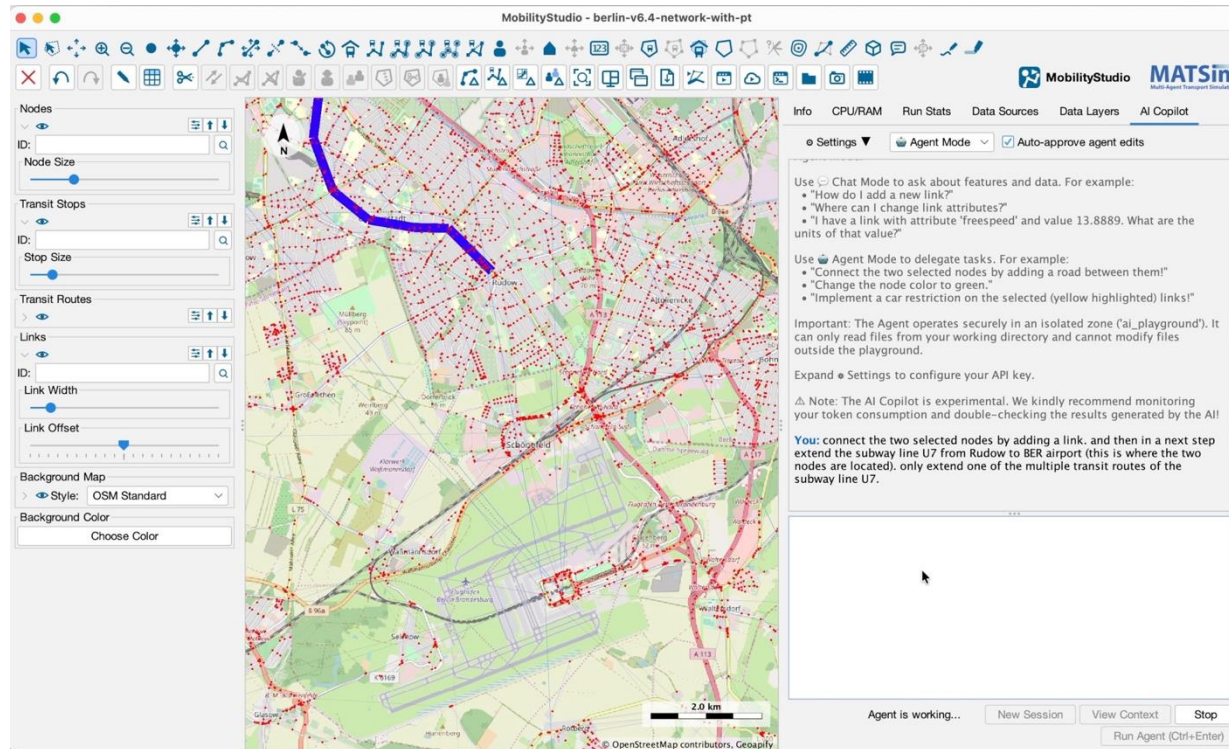
Run MATSim with a single click. Visualize and analyze the simulation results.



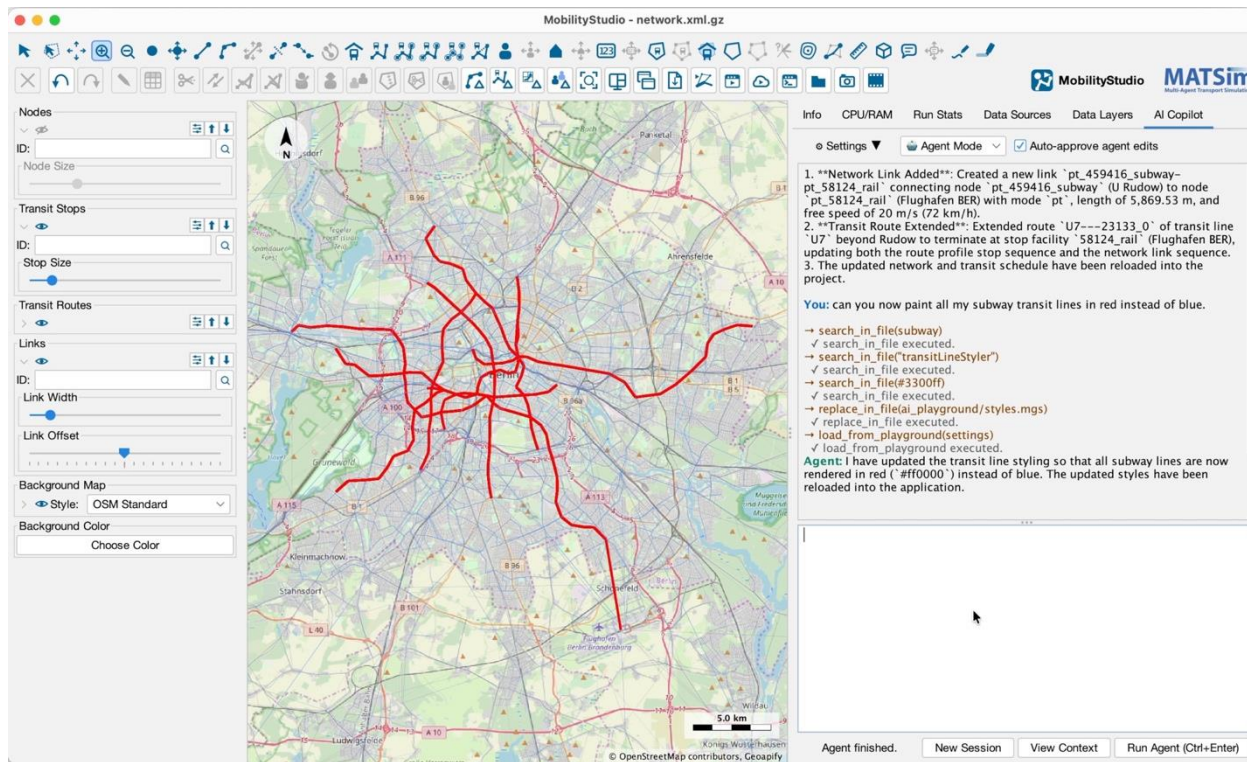
Could we have asked the AI to do that?

The integrated AI Copilot in MobilityStudio...

Using the integrated AI Copilot to extend the U7 subway line to BER airport



Using the AI to adjust graphical settings in MobilityStudio.



MobilityStudio - network.xml.gz

Nodes
ID:
Node Size:

Transit Stops
ID:
Stop Size:

Transit Routes
ID:
Link Width:
Link Offset:

Background Map
Style: OSM Standard
Background Color: Choose Color

Info CPU/RAM Run Stats Data Sources Data Layers AI Copilot

Settings Agent Mode Auto-approve agent edits

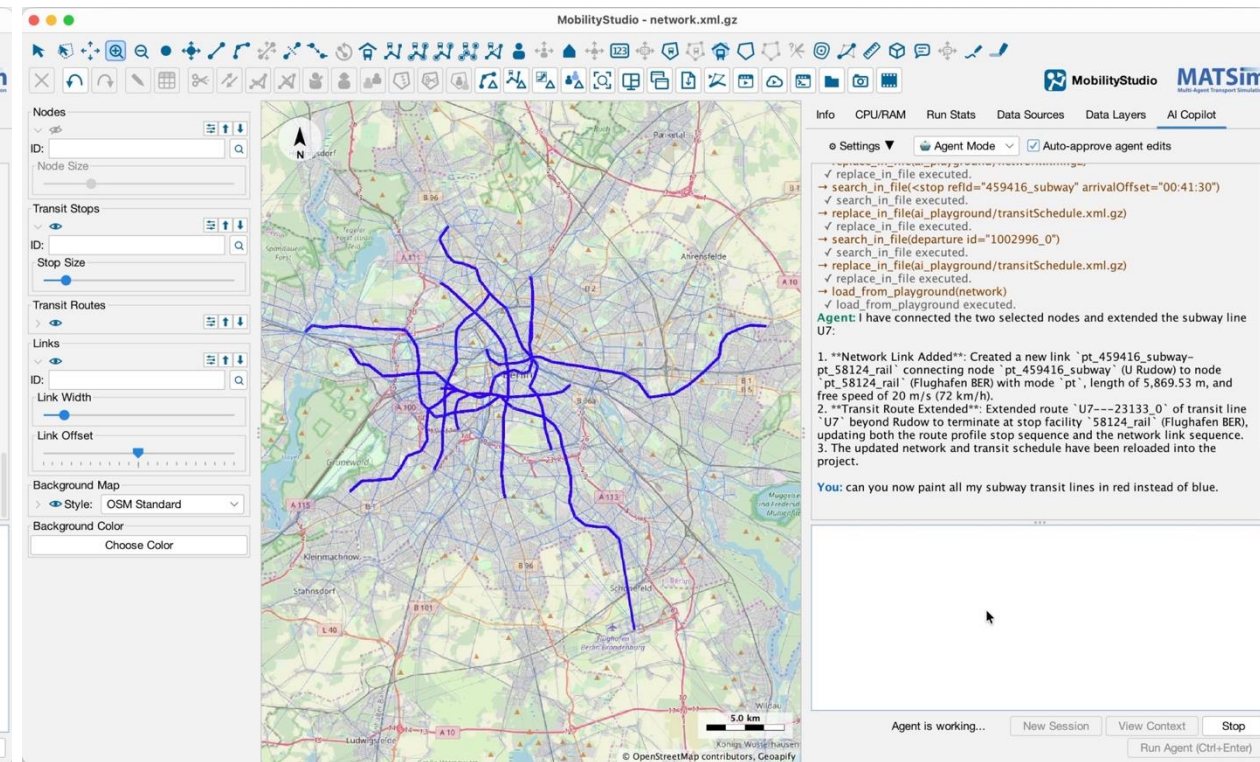
1. ****Network Link Added****: Created a new link 'pt_459416_subway-pt_58124_rail' connecting node 'pt_459416_subway' (U Rudow) to node 'pt_58124_rail' (Flughafen BER) with mode 'pt', length of 5,869.53 m, and free speed of 20 m/s (72 km/h).
2. ****Transit Route Extended****: Extended route 'U7---23133_0' of transit line 'U7' beyond Rudow to terminate at stop facility '58124_rail' (Flughafen BER), updating both the route profile stop sequence and the network link sequence.
3. The updated network and transit schedule have been reloaded into the project.

You: can you now paint all my subway transit lines in red instead of blue.

→ search_in_file(subway)
✓ search_in_file executed.
→ search_in_file(transitLineStyle)
✓ search_in_file executed.
→ search_in_file(#3300ff)
✓ search_in_file executed.
→ replace_in_file(ai_playground/styles.mgs)
✓ replace_in_file executed.
→ load_from_playground(settings)
✓ load_from_playground executed.

Agent: I have updated the transit line styling so that all subway lines are now rendered in red ('#ff0000') instead of blue. The updated styles have been reloaded into the application.

Agent finished. New Session View Context Run Agent (Ctrl+Enter)



MobilityStudio - network.xml.gz

Nodes
ID:
Node Size:

Transit Stops
ID:
Stop Size:

Transit Routes
ID:
Link Width:
Link Offset:

Background Map
Style: OSM Standard
Background Color: Choose Color

Info CPU/RAM Run Stats Data Sources Data Layers AI Copilot

Settings Agent Mode Auto-approve agent edits

✓ replace_in_file executed.
→ search_in_file(<stop refId="459416_subway" arrivalOffset="00:41:30")
✓ search_in_file executed.
→ replace_in_file(ai_playground/transitSchedule.xml.gz)
✓ replace_in_file executed.
→ search_in_file(departure id="1002996_0")
✓ search_in_file executed.
→ replace_in_file(ai_playground/transitSchedule.xml.gz)
✓ replace_in_file executed.
→ load_from_playground(network)
✓ load_from_playground executed.

Agent: I have connected the two selected nodes and extended the subway line U7:

1. ****Network Link Added****: Created a new link 'pt_459416_subway-pt_58124_rail' connecting node 'pt_459416_subway' (U Rudow) to node 'pt_58124_rail' (Flughafen BER) with mode 'pt', length of 5,869.53 m, and free speed of 20 m/s (72 km/h).
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3. The updated network and transit schedule have been reloaded into the project.

You: can you now paint all my subway transit lines in red instead of blue.

Agent is working... New Session View Context Stop
Run Agent (Ctrl+Enter)

AI Usecase: Building scenarios

- We get what we get: a scanned overview map and a table from a planning report
- The problem: Planning documents don't speak MATSim
- Let's see what AI can do!*

1 Leerer Landstraße B72 Anbindung, Ortsteil Schirum

Nr.	Wohngebiet	WE	ha	Stand	Format
349	Stiegelhörner Weg	26	2,0	Rechtsverb.	pdf
Satz 63	Ortskern Schirum	22	18,4	Aufstellung	pdf

langfristige Siedlungsentwicklung

	Fläche	WE	ha	SEK
1	Fläche	10	1,1	SEK
2	Fläche	20	2,6	SEK

2 Kirchdorfer Straße K111 Anbindung, Ortsteil Kirchdorf

Nr.	Wohngebiet	WE	ha	Stand	Format
389	Weizenstraße	30	3,8	Auslegung	pdf

3 Oldersumer Straße L1 Anbindung, Ortsteil Rahe

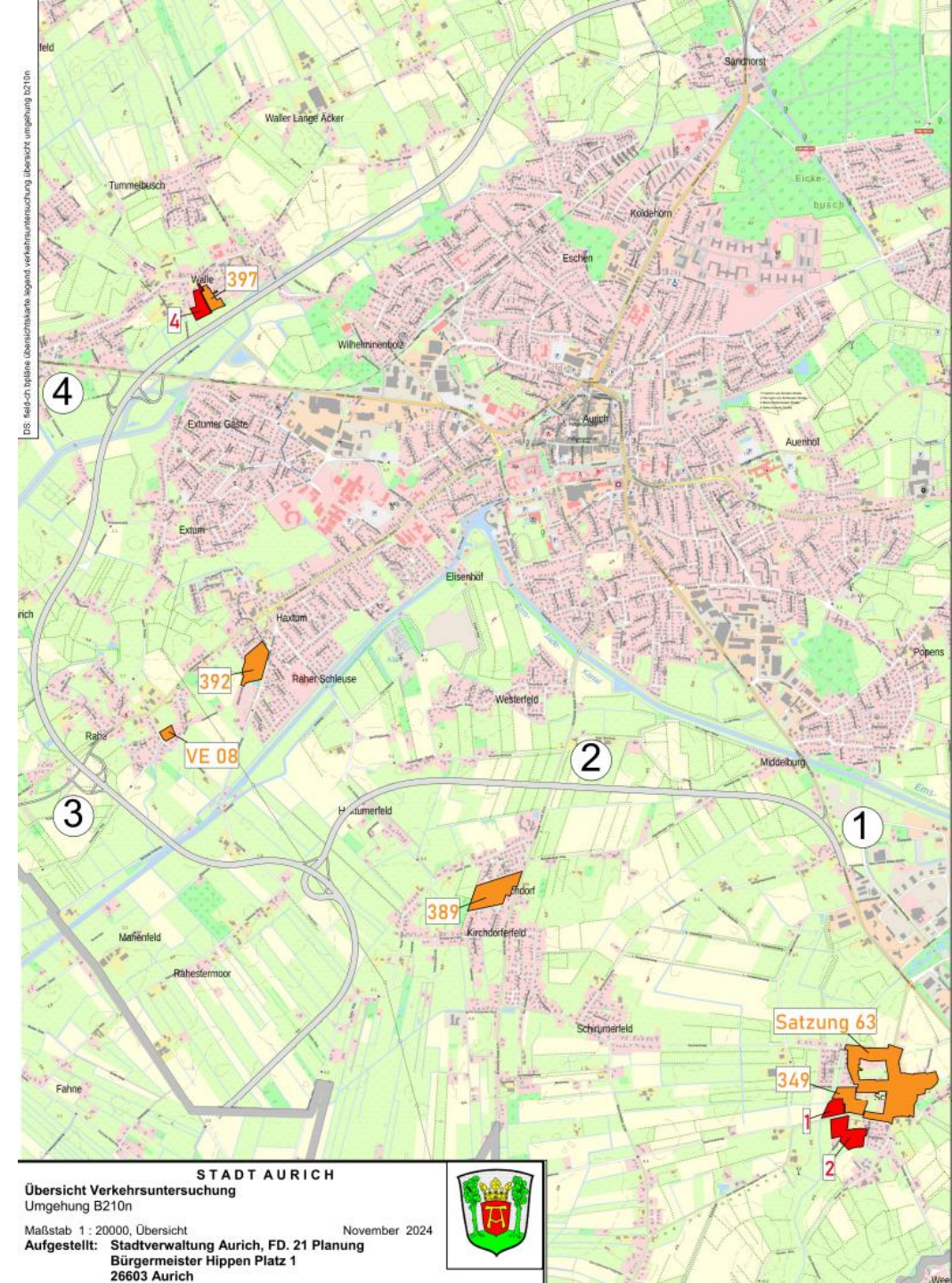
Nr.	Wohngebiet	WE	ha	Stand	Format
VE08	Rahe, Boomweg	20	5,5	Aufstellung	pdf
392	Hinter der Mühle	40	2,6	Aufstellung	pdf

4 Emder Straße B210 Anbindung, Ortsteil Walle

Nr.	Wohngebiet	WE	ha	Stand	Format
397	Südl. Binnen Diehl	12	1,1	Aufstellung	pdf

langfristige Siedlungsentwicklung

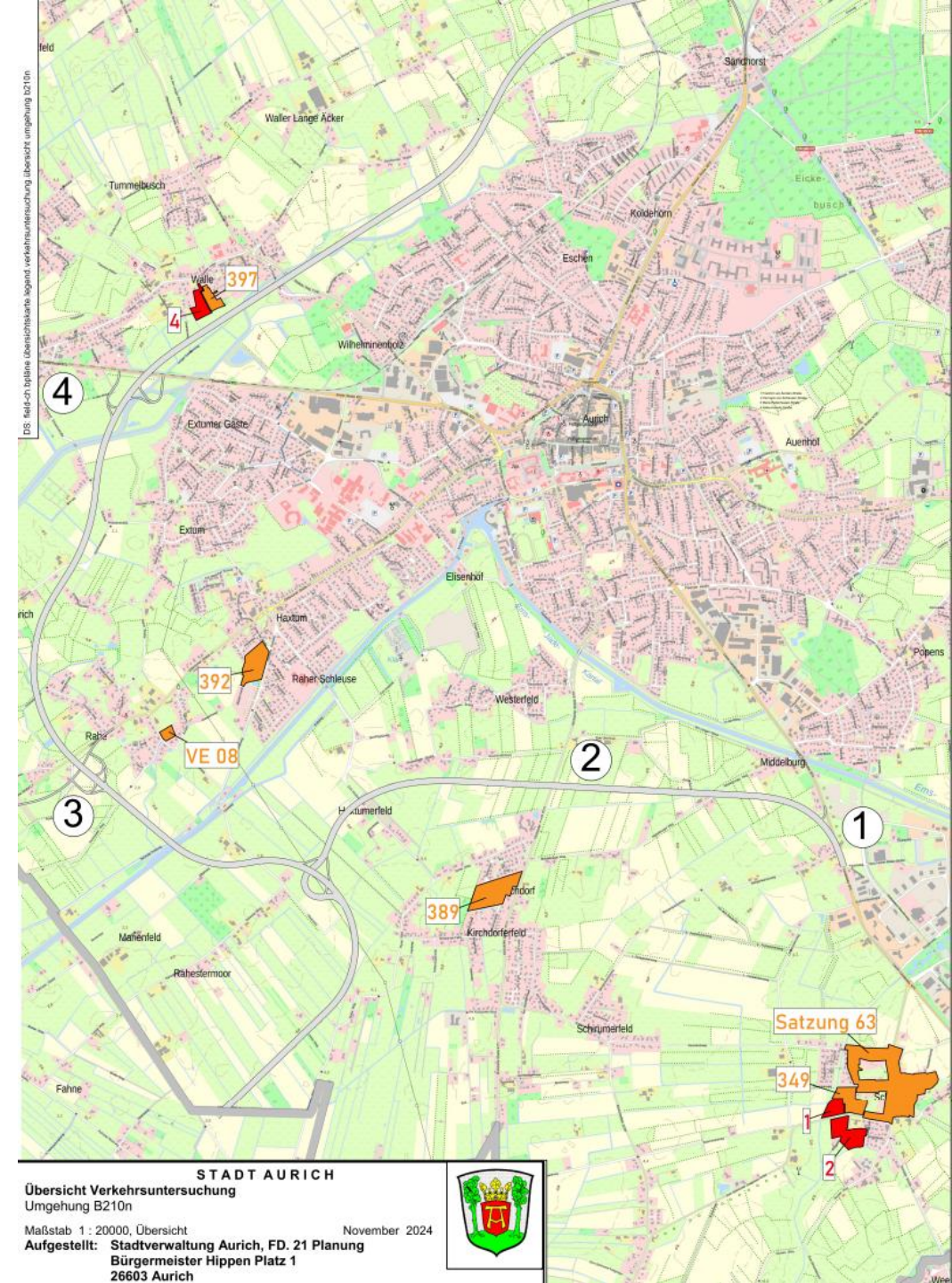
	Fläche	WE	ha	SEK
4	Fläche	10	1,3	SEK



*In the actual project this was done without AI

Building the Bypass B210n

“In this folder you see a MATSim network file and a pdf with a picture in it. The pdf depicts the area around the German town Aurich including a future road called B210n (in grey). The MATSim network file is of an area larger than the pdf file but fully includes it. The coordinates of the file are in EPSG:25832. By comparing the pdf picture to OpenStreetMap find out what the coordinates of the course of the B210n are as well as its connectors and insert it into the network file. Assume car,ride as the allowed modes and find a suitable freespeed yourself using other larger roads in the network as reference. If you find additional info about the B210n on the web you may use that too (all the info will probably be in German).”



Building the Bypass B210n

Georeference the image

Match landmarks and the printed scale to real-world coordinates (EPSG:25832) — the map has no embedded geodata of its own.

Isolate the B210n

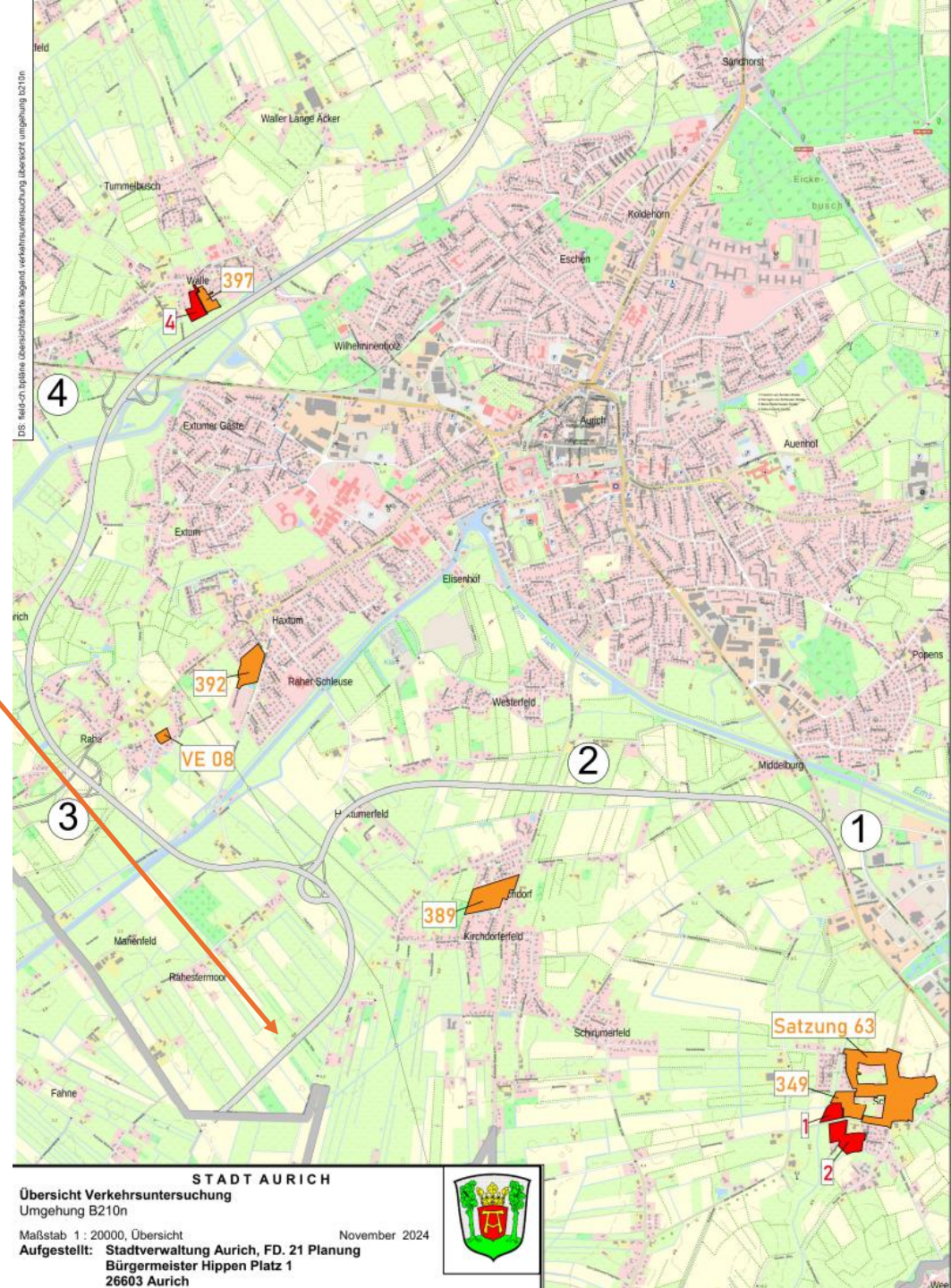
Figure out via online research that the south arm is not part of what we want to include, since it will be built in the very far future under a different name.

Choose realistic params

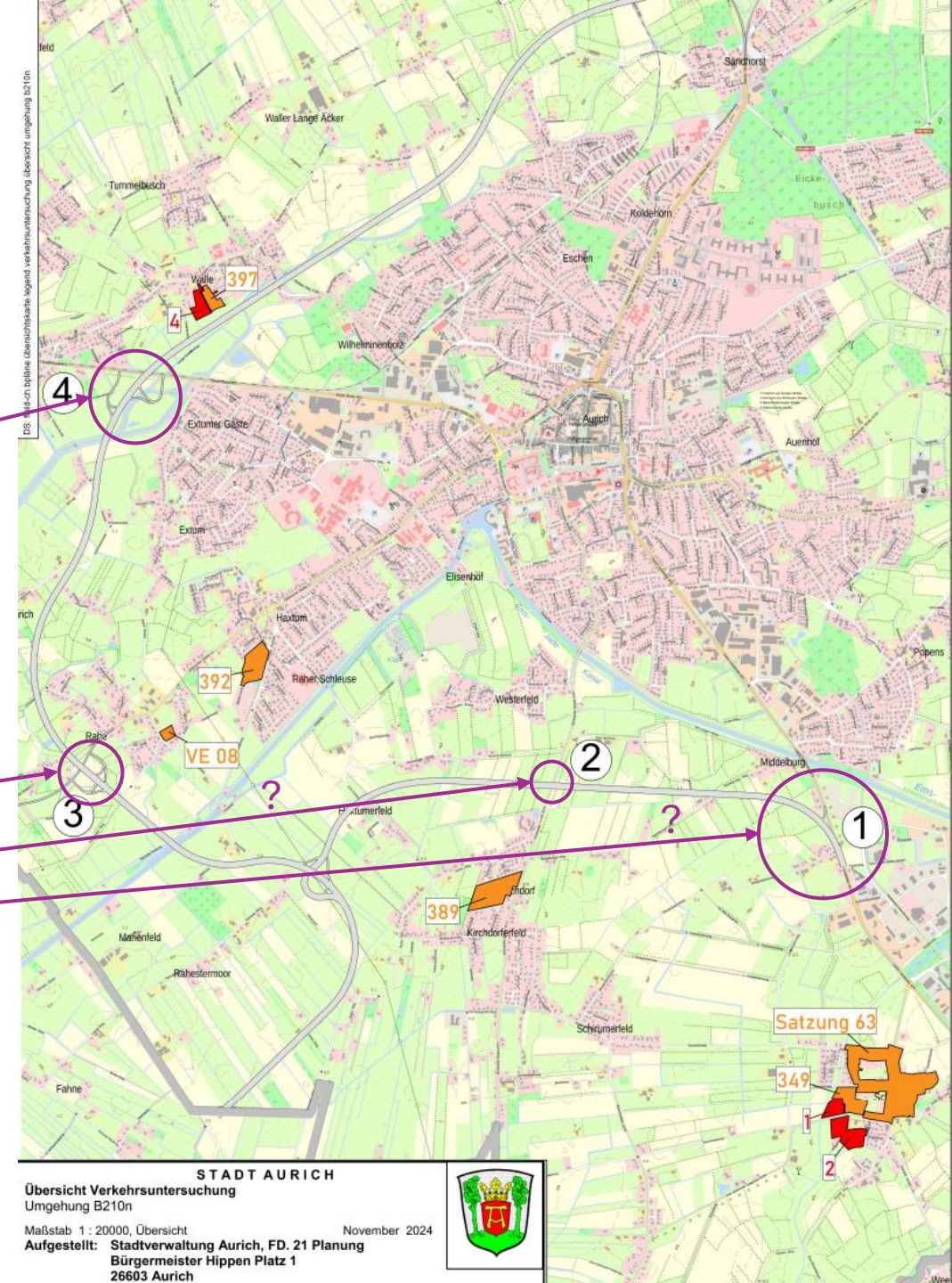
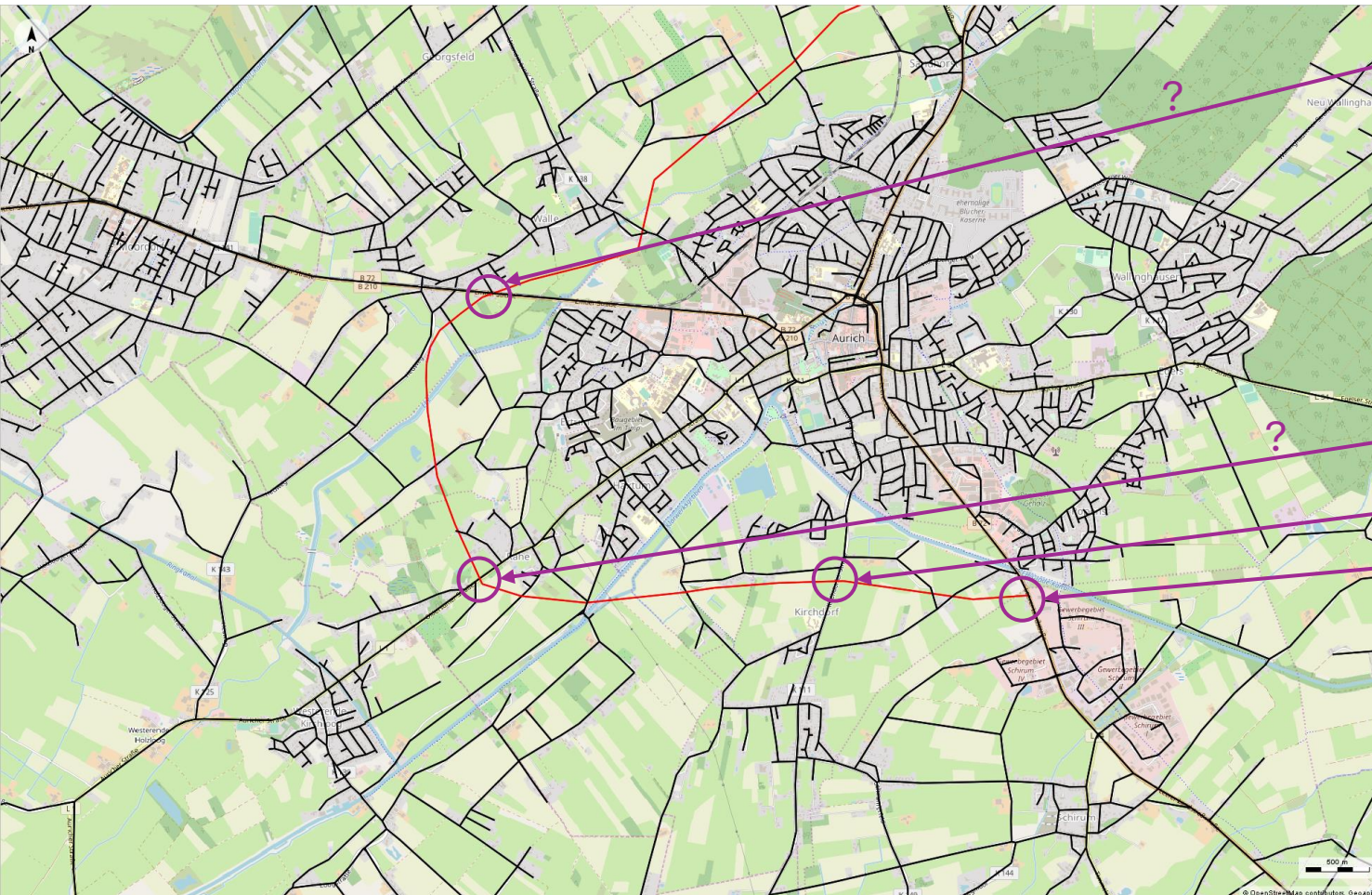
Set freespeed and capacity by comparison with similar classified roads already coded in the network — B-road vs. residential vs. ramp.

Trace the alignment

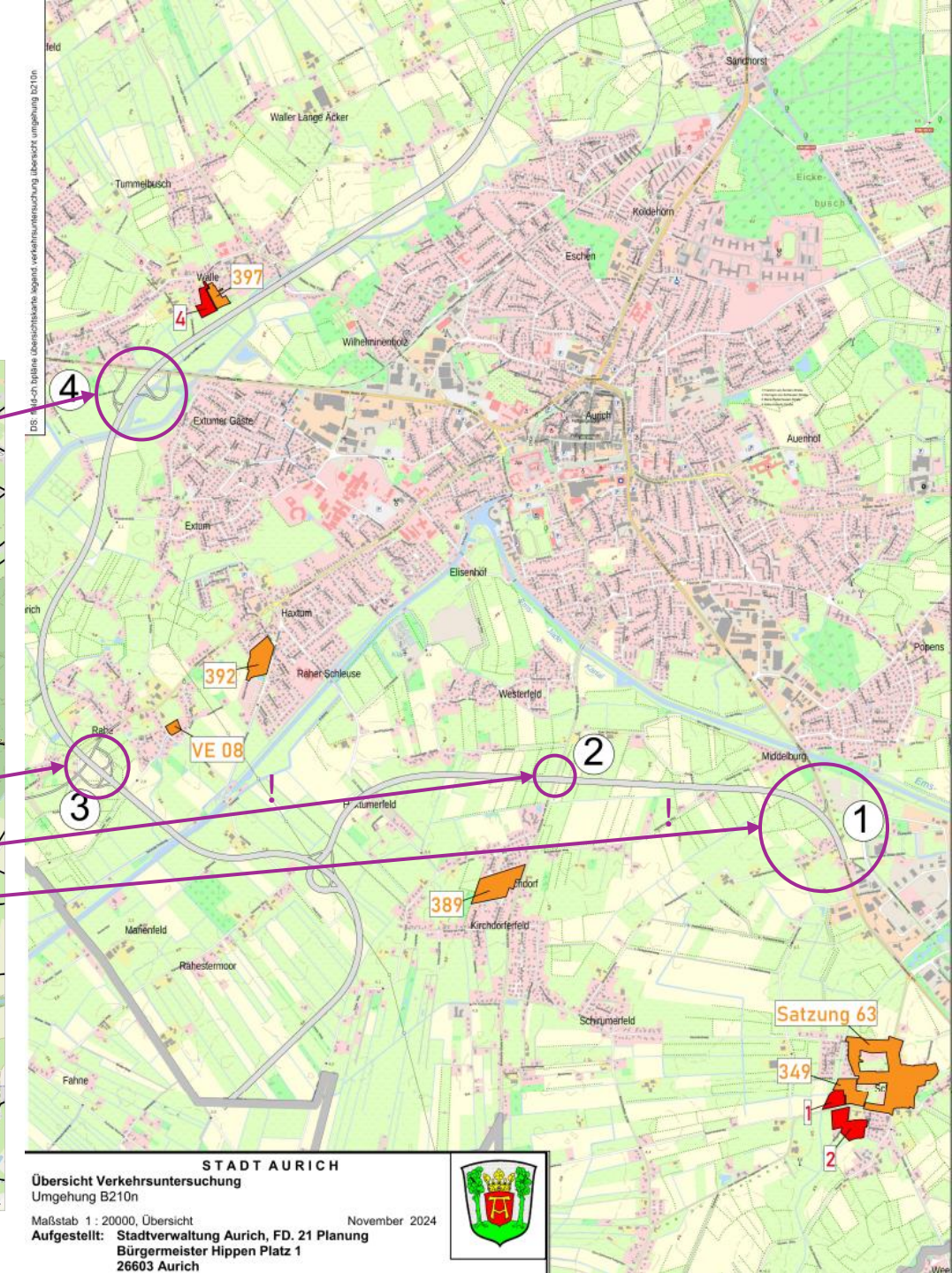
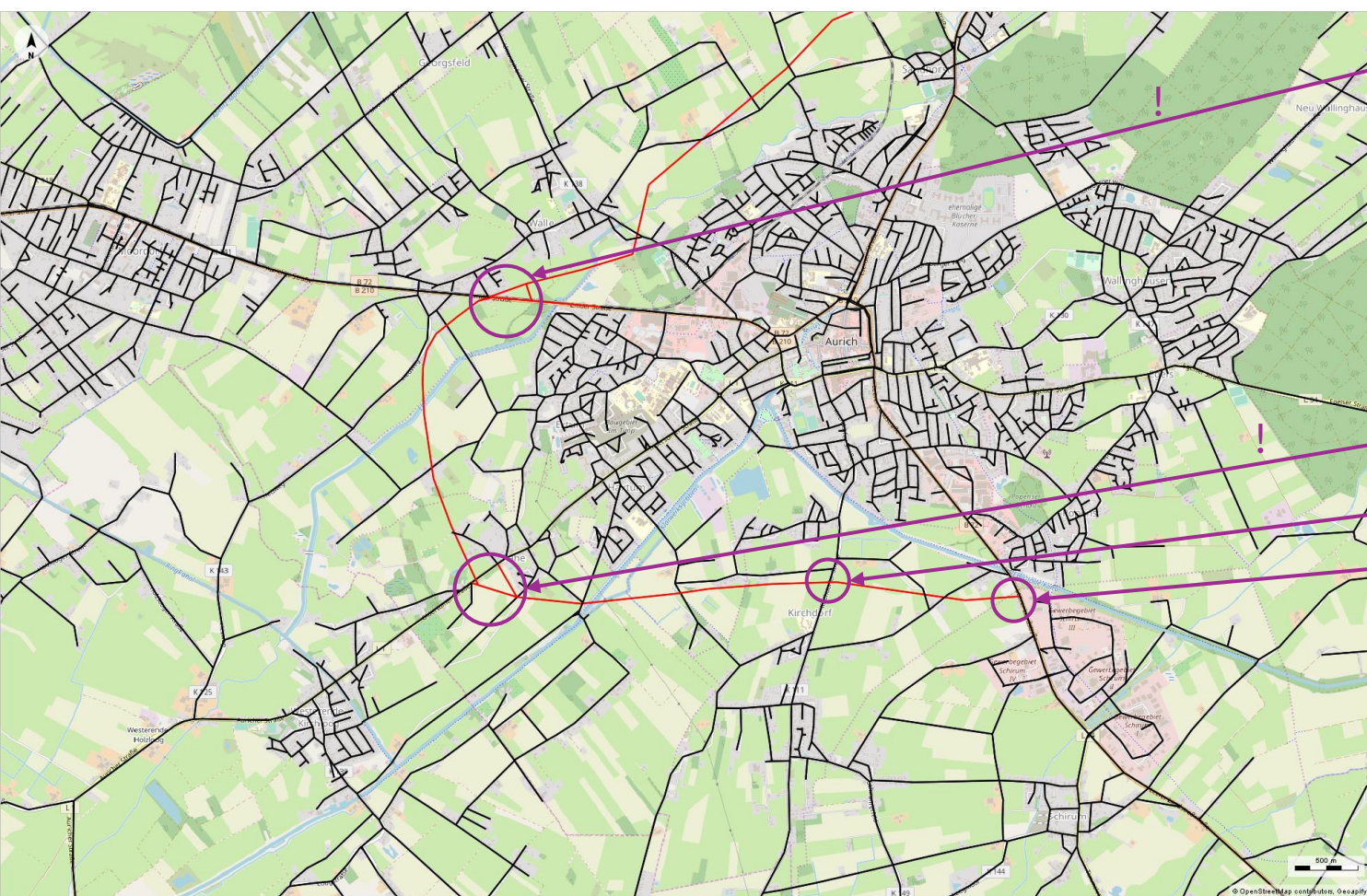
Follow the drawn route bend by bend, cross-checking every curve and junction against OpenStreetMap's existing road pattern for the same streets.



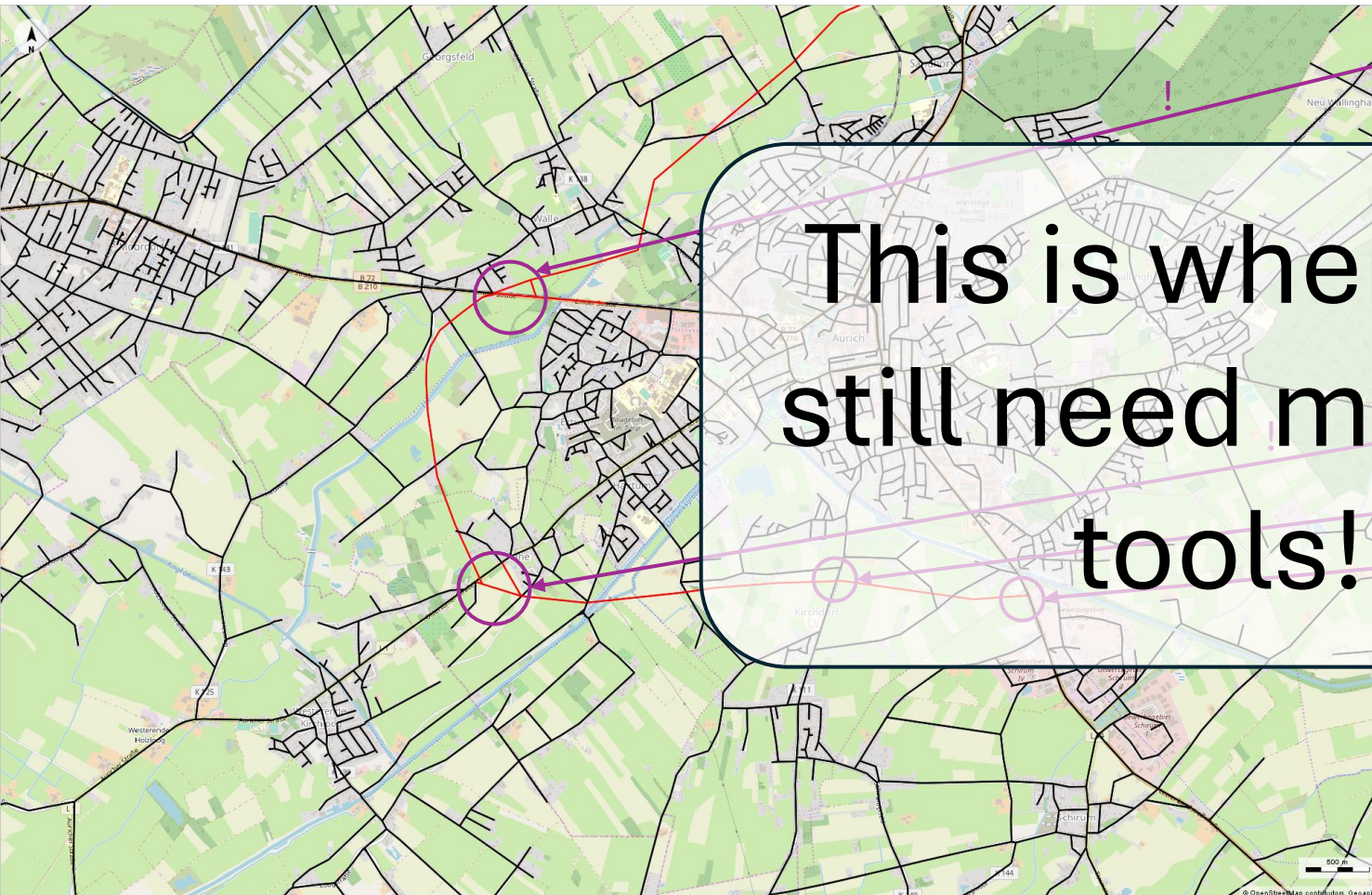
B210n – First Shot



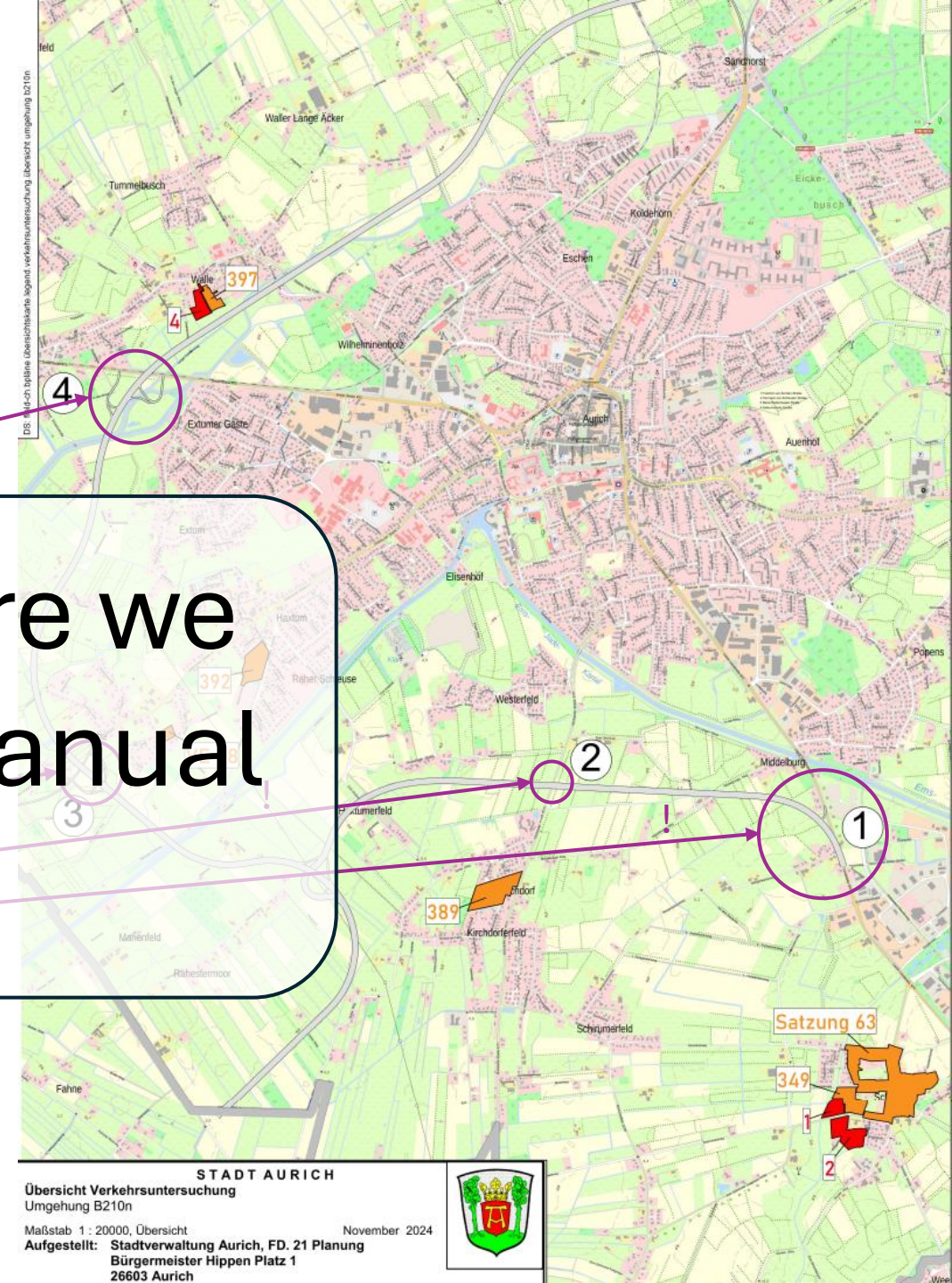
After Some Iterations



After Some Iterations



This is where we
still need manual
tools!



New Home Facilities

Parse the pdf table

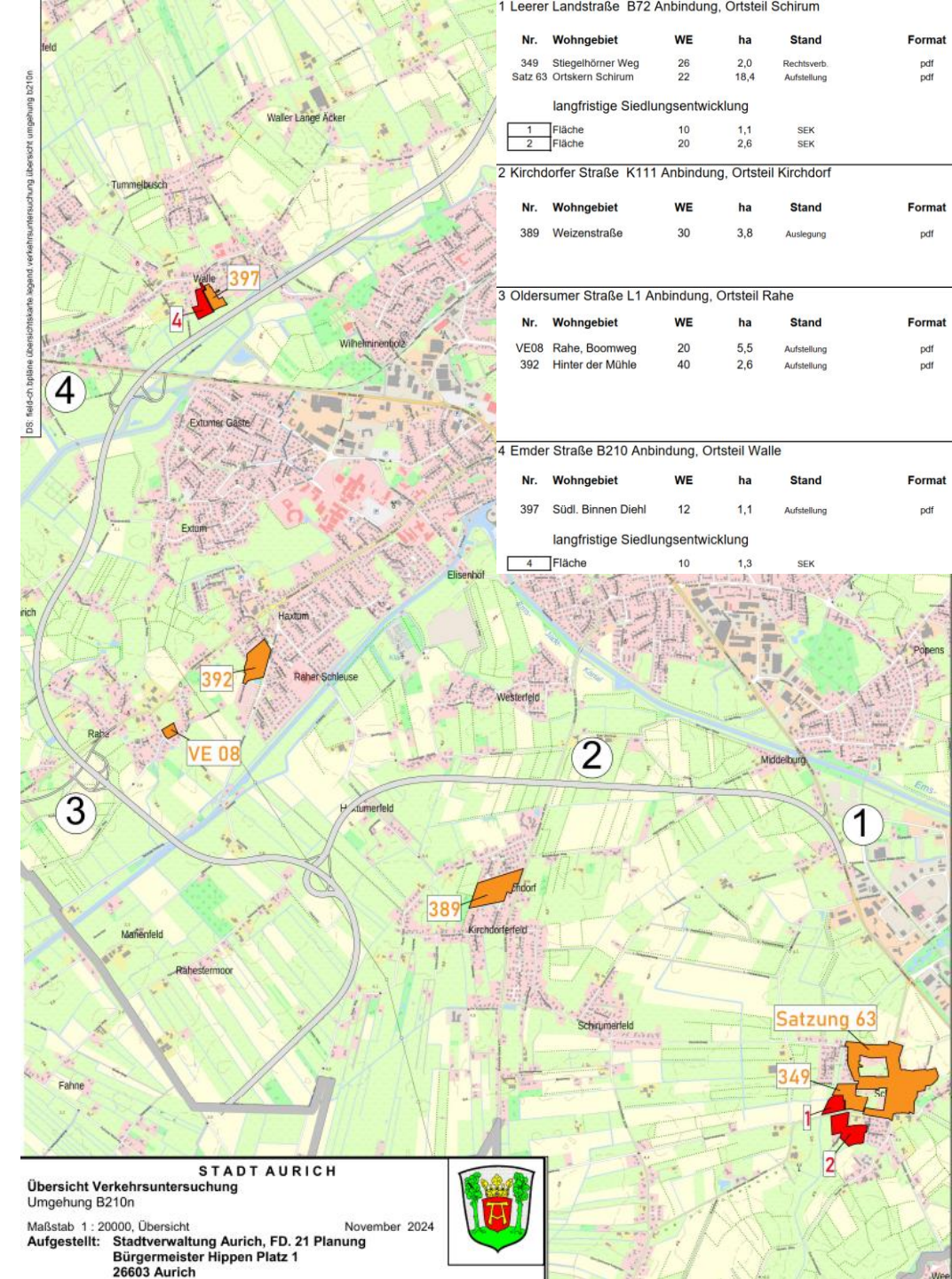
Find out how many flats (“WE”) will be located in each area.

Isolate the orange areas

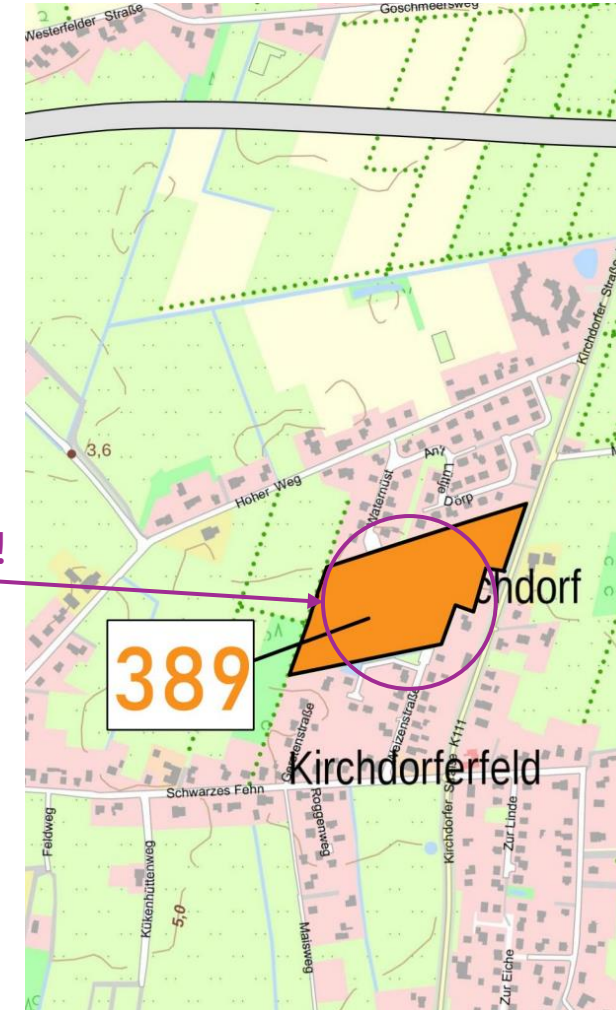
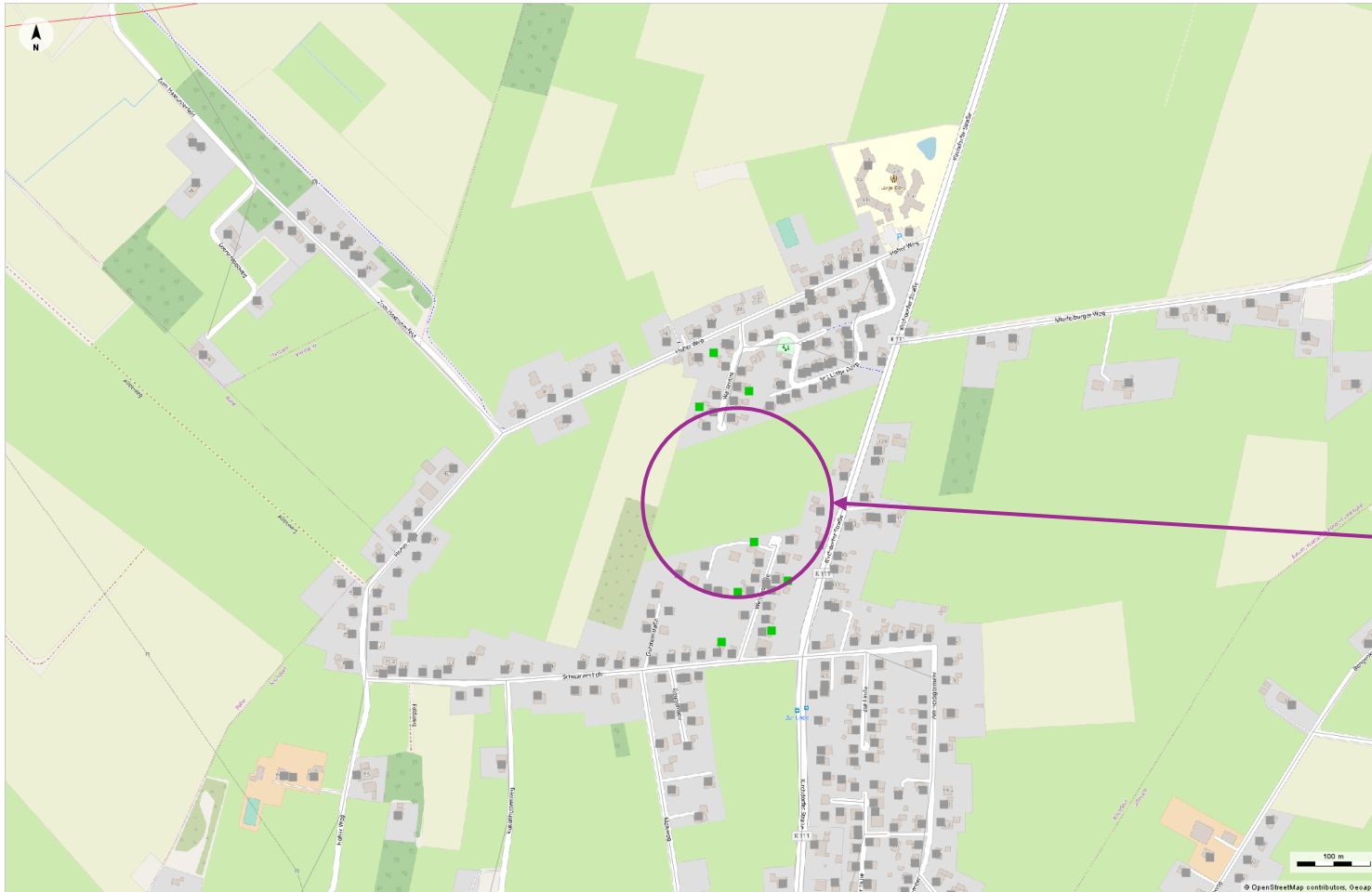
Geolocate the development areas and parse their labels and cross reference them with the table.

Place facilities

Place the correct number of facilities with reasonable capacities in the correct areas.



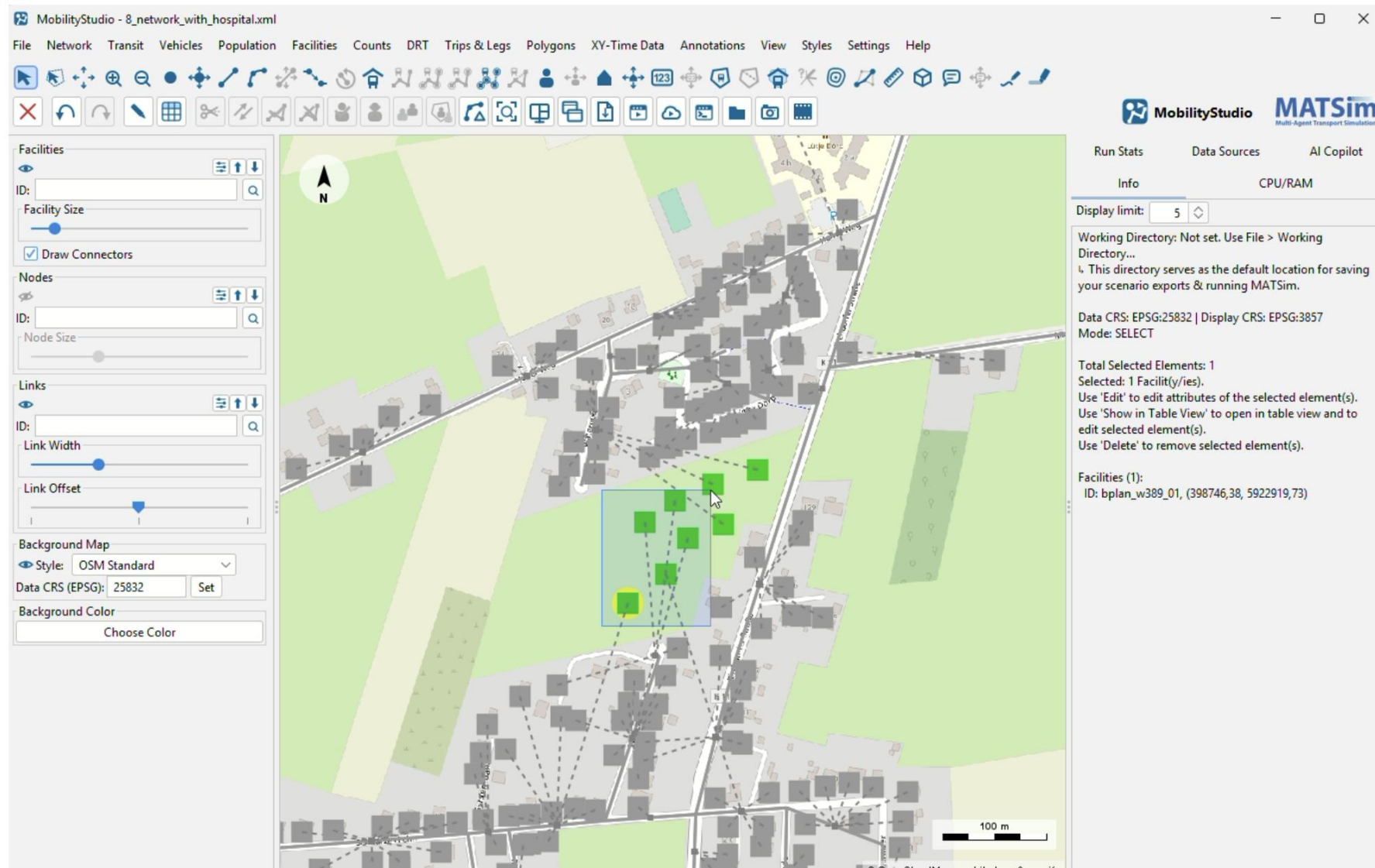
New Home Facilities



New Home Facilities



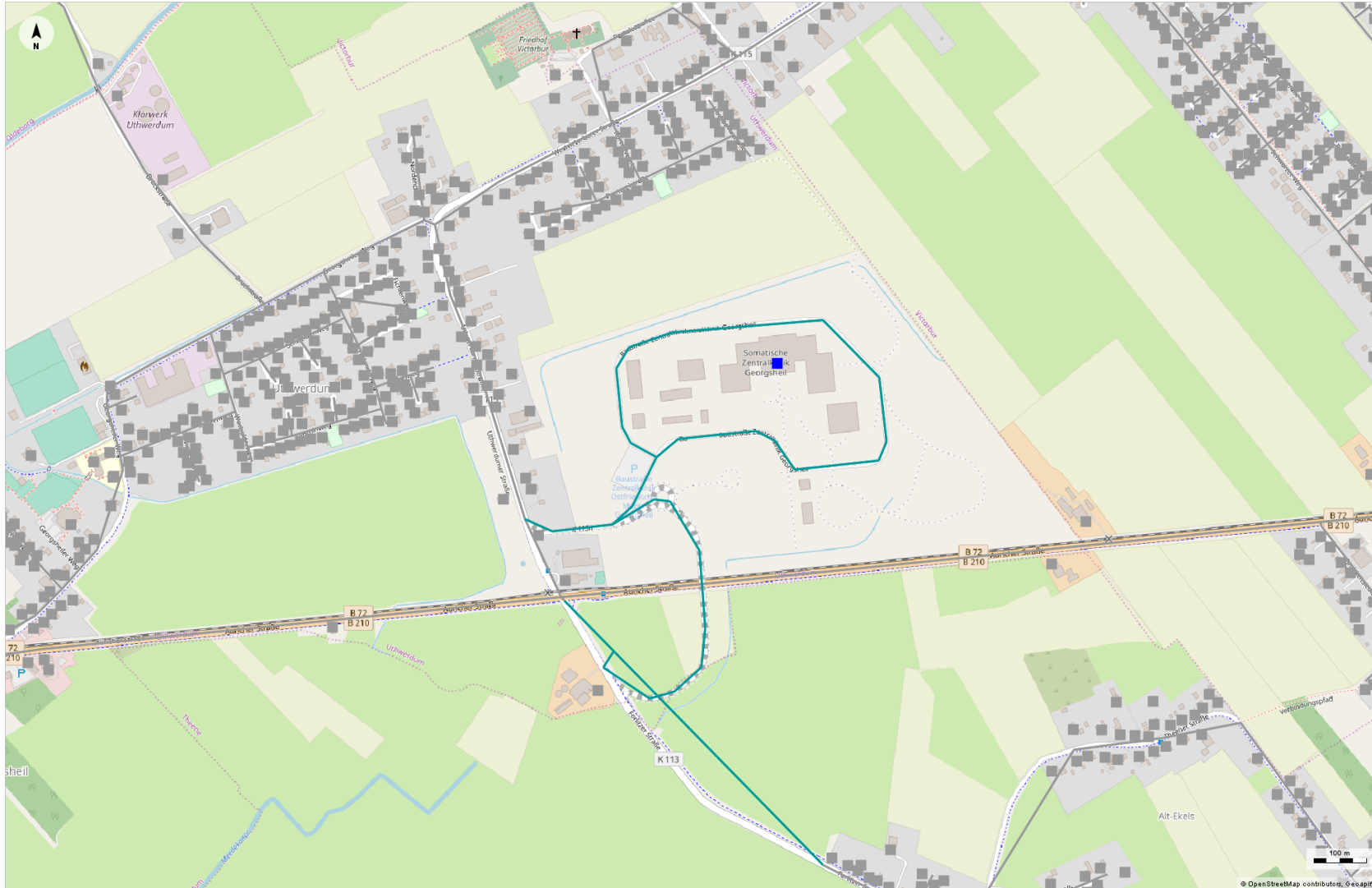
Fixing the Home Facilities



Final Test: Add something without documents

“do some online research about the "Zentralklinikum Georgsheil" (alternative name "Zentralklinik Ostfriesische Meere"). Place a facility where it is going to be. The facility should have [redacted] work capacity and an errand capacity of about 10% of all other errand capacities combined. Make new streets connecting the facility to the existing network. The planned layout of those streets is already in OpenStreetMap but you may also use other online sources. This is a complicated task. If you are unsure about something feel free to ask me.”

Final Test: Add something without documents



AI experiment takeaways written by AI:

TAKEAWAYS

What this changes — and what it doesn't

GOOD FIT FOR AI ASSISTANCE



Digitizing scanned plans into georeferenced network edits



Cross-referencing documents against a live map, at scale



Proportional / quantitative allocation across many zones



A reviewable first-pass extension, not a final answer

STILL NEEDS A DOMAIN EXPERT



Judging whether an interchange geometry is realistic



Knowing what a real junction or tunnel should look like



Deciding what belongs in scope for this scenario

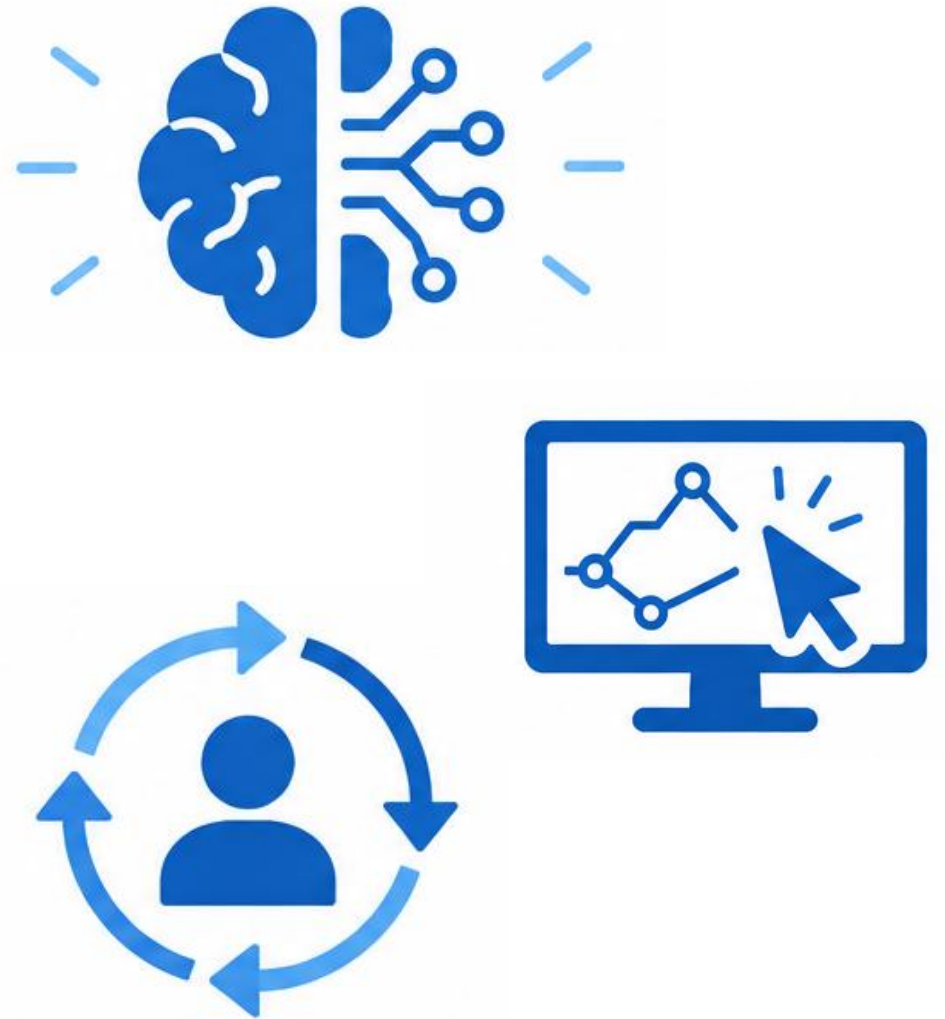


Final sign-off before the scenario is used for analysis

Every correction in this walkthrough came from a reviewer who knew exactly what to look for — that part doesn't change.

Conclusion

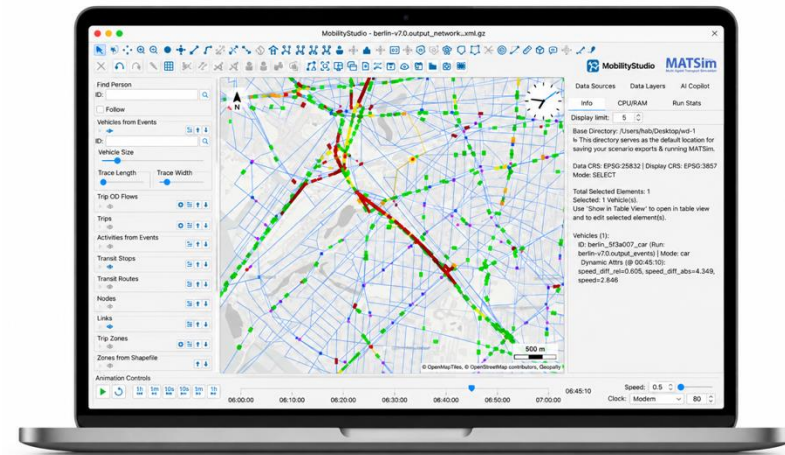
- MATSim is powerful (enough). Accessibility is the challenge...
- Much of the effort happens before and after the simulation. A graphical, integrated environment can improve existing MATSim workflows and lower the barrier for new users.
- AI is changing how we work with MATSim. But even with AI-assisted or agentic workflows, visual tools remain crucial:
 - to see, test and validate what AI did
 - to make precise edits intuitively (rather than telling the agent to do so)
 - to keep humans in the loop
- Both AI and intuitive tools like MobilityStudio can work together to make MATSim accessible to a much broader group of users.



Thank you!

Want to download and try MobilityStudio?

www.mobilitystudio.de



Need an application-ready MATSim model?

www.senozon.com

senozon

